

Assignment: Detect Intent-to-Speak from EEG

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1 Problem Setup

The objective is to detect whether a participant is about to speak using EEG signals recorded from a small number of electrodes. This is formulated as a binary classification problem, where each EEG sample is assigned one of two labels:

$$y = \begin{cases} 1, & \text{intent to speak} \\ 0, & \text{no intent to speak} \end{cases}$$

Let t_0 denote the speech onset time of the participant. A positive sample is defined as the EEG interval ending exactly at the participant's speech onset:

$$X^+ = EEG[t_0 - 500 \text{ ms}, t_0]$$

This window represents the 500 ms period immediately before speech begins and is labeled as intent-to-speak.

All other EEG windows whose interval end is not aligned with the participant's speech onset are labeled as negative samples:

$$X^- = EEG[t - 500 \text{ ms}, t], \quad t \neq t_0$$

where t is not a speech onset timestamp. Negative samples will be drawn from silent, idle, listening, and non-speaking periods.

1.1 Obtaining Speech Onset Ground Truth

Speech onset is required as the ground truth reference for labeling positive EEG windows. The preferred setup is to record EEG and microphone audio using the same acquisition device or the same hardware clock. In this case, the microphone signal can be recorded as an auxiliary channel along with the EEG. This is the most accurate synchronization method because the EEG and audio signals are sampled on the same time axis.

If the EEG device does not support direct microphone input, EEG and microphone audio will be synchronized using Lab Streaming Layer (LSL) and LabRecorder. In this setup, the EEG stream and microphone stream are recorded simultaneously through LSL, and LabRecorder stores them in a single synchronized XDF file. LSL provides timestamps for each stream and allows the EEG and audio signals to be aligned during offline analysis.

Speech onset will be detected from the synchronized microphone waveform. Specifically, the onset time t_0 will be estimated as the first sustained increase in short-time audio energy above the pre-speech baseline. These automatically detected onsets may be visually inspected and corrected if necessary. Once t_0 is obtained, the corresponding EEG window from $t_0 - 500$ ms to t_0 will be extracted and labeled as a positive intent-to-speak sample.

2 Data Collection Plan

Data will be collected from approximately 20 healthy adult participants. Participants will be selected to include variability in speaking habits and conversational styles, such as frequent speakers, less frequent speakers, people who are comfortable speaking in conversations, and people who speak less spontaneously. Where possible, diversity in age and gender will also be included. This variability is important because intent-to-speak EEG patterns may differ across individuals.

Each participant will complete approximately 100–150 speech trials. With 20 participants, this would provide around 2000–3000 positive speech-onset examples. Additional negative samples will be extracted from silent, idle, and listening periods. Since no-intent periods are naturally more common than speech-onset events, negative samples will be selected carefully rather than using all possible negative windows.

The data collection protocol will include both controlled and semi-naturalistic conditions. Controlled conditions will be used for the main training data because they provide cleaner speech-onset labels. Semi-naturalistic conditions will be included mainly to test robustness in more realistic environments.

2.1 Experimental Conditions

2.1.1 Read-Aloud Task

In the read-aloud condition, the participant will read visually presented words or short sentences aloud. For example, a word or sentence may be shown on a screen, and the participant will read it aloud naturally. This condition provides clean and repeatable speech-onset events, making it suitable for the primary training data.

2.1.2 Question-Answer Task

In the question-answer condition, the participant will respond naturally to simple prompts or questions. For example, the participant may be asked a question such as “What did you eat for breakfast?” and will answer in their own words. This condition is more natural than reading aloud while still preserving a relatively controlled structure.

2.1.3 One-on-One Conversation Task

In the one-on-one conversation condition, the participant will have a short conversation with the experimenter. This condition is included to capture more natural speech preparation and turn-taking behavior. Unlike the read-aloud task, the timing of speech is less strictly controlled, but the participant’s speech onset can still be detected from the microphone signal.

2.1.4 Non-Speaking and Listening Conditions

Non-speaking conditions will be recorded to obtain negative no-intent samples. These conditions will include sitting idle, listening to another person speak, watching or listening without responding, and being in the same room as another person without actively speaking. These periods are important because the model should learn to distinguish actual intent-to-speak from silence, listening, and passive social presence.

2.2 Semi-Naturalistic Robustness Condition

A small semi-naturalistic condition may also be included, such as a group discussion or meeting-like setting with two or three people. In this condition, the EEG-wearing participant may sometimes speak and sometimes listen silently. This condition resembles realistic use cases, but it is harder to label accurately because multiple speakers may be present. Therefore, this condition will mainly be used for robustness testing rather than primary model training.

In group or meeting-like settings, a close-talk microphone will be placed near the EEG-wearing participant, or separate microphones will be used for each speaker. This is necessary to ensure that only the participant’s own speech onset is used as a positive label. Speech from other people will not be treated as the participant’s speech onset.

2.3 Assumptions

The main assumption is that only the participant wearing the EEG electrodes is being labeled. If another person speaks while the participant is listening, that period will be treated as a no-intent or listening condition for the participant, unless the participant also begins speaking. Another assumption is that speech onset can be detected reliably from the synchronized microphone recording. The controlled tasks are expected to provide cleaner labels, while the conversation and meeting-like tasks are expected to be noisier but more realistic.

3 Hardware and Electrode Choice

EEG will be recorded using a portable research-grade EEG acquisition system with support for 2–4 active electrodes. A research-grade system is preferred over a consumer headset because the pre-speech EEG signal is expected to be weak, and reliable low-noise recording is important when using only a small number of electrodes. The EEG device should support data streaming through Lab Streaming Layer (LSL), so that the EEG stream and microphone stream can be synchronized and recorded together using LabRecorder.

3.1 Sampling Rate

EEG will be recorded at a sampling rate of 250 Hz. This provides a temporal resolution of 4 ms and gives 125 samples in each 500 ms pre-speech window:

$$0.5 \text{ s} \times 250 \text{ Hz} = 125 \text{ samples}$$

This is sufficient for analyzing EEG activity related to speech preparation, which is expected to lie mainly in low and medium frequency ranges such as theta, mu/alpha, and beta activity. Since the main analysis focuses on EEG activity below approximately 40–45 Hz, a 250 Hz sampling rate is adequate according to the Nyquist principle.

If the EEG device records at a higher native sampling rate, such as 500 Hz, the data may be downsampled to 250 Hz after preprocessing. Downsampling means reducing the number of samples per second. For example, a 500 Hz signal has 250 samples in a 500 ms window, while a 250 Hz signal has 125 samples in the same window. Reducing the sampling rate decreases data size, computational cost, and model complexity while preserving the EEG information needed for this task.

Downsampling should be performed only after low-pass or bandpass filtering. This is important because high-frequency noise can otherwise be misrepresented as false lower-frequency activity after downsampling, a problem known as aliasing. Therefore, the EEG will first be filtered to remove unwanted high-frequency components, and only then will downsampling be applied if necessary.

3.2 Filtering Considerations

Before feature extraction or modeling, the EEG will be bandpass filtered, for example from 0.5–45 Hz. The lower cutoff removes slow baseline drift, while the upper cutoff removes high-frequency noise and muscle artifacts that are less relevant for the intended EEG analysis. Since recordings are expected to be collected in India, a 50 Hz notch filter may also be applied to reduce power-line interference if needed.

The filtering step is especially important before any downsampling. If high-frequency noise remains in the signal and the sampling rate is reduced, that noise may fold into the lower-frequency range and appear as artificial EEG activity. Filtering before downsampling prevents this and ensures that the lower sampling rate still represents the relevant EEG content accurately.

3.3 Electrode Placement

The preferred four-electrode montage will use FCz, Cz, C3, and C4 according to the international 10–20 EEG system. These electrodes are chosen because intent-to-speak is expected to involve speech planning and motor preparation before actual speech onset. Speaking requires coordinated movement of the lips, tongue, jaw, vocal tract, and breathing muscles. Therefore, fronto-central and central sensorimotor regions are more relevant than posterior visual regions for this task.

FCz is placed over the fronto-central midline region. This location is included because it may capture premotor and action-preparation activity before speech begins. Since the goal is to detect intent before overt speech onset, activity related to planning and preparation is particularly important.

Cz is placed over the central midline region. This location is useful for capturing general motor preparation. Since speaking is a motor action involving articulatory muscles, Cz is expected to provide information about the preparation of movement before speech.

C3 and C4 are placed over the left and right central regions, respectively. These locations provide bilateral sensorimotor coverage. C3 may capture left central activity related to speech and motor preparation, while C4 provides the corresponding right central coverage. Including both channels helps capture bilateral movement-related activity and reduces the risk of relying on only one side of the scalp.

Thus, the preferred electrode set is:

$$\{FCz, Cz, C3, C4\}$$

If only two electrodes are available, FCz and Cz will be used because they provide the most direct coverage of fronto-central planning and central motor-preparation regions. If three electrodes are available, FCz, Cz, and C3 will be used. With four electrodes, FCz, Cz, C3, and C4 will be used as the preferred montage.

Reference and ground electrodes will be placed according to the EEG device manufacturer’s recommended configuration, for example using an earlobe or mastoid reference and a forehead or AFz ground. These reference and ground electrodes are required for stable EEG recording but are not counted as the main 2–4 signal electrodes.

4 Preprocessing

The EEG and microphone streams will first be loaded from the synchronized recording. If Lab Streaming Layer and LabRecorder are used, both streams will be available in a single synchronized XDF file. Speech onset times will be obtained from the microphone waveform and used as reference points for extracting EEG windows.

4.1 Filtering

The raw EEG will be bandpass filtered from 0.5–45 Hz. The lower cutoff removes slow baseline drift, while the upper cutoff removes high-frequency noise that is less relevant for speech-preparation EEG analysis. Since the recordings are expected to be collected in India, a 50 Hz notch filter may also be applied if power-line interference is visible in the signal.

If the EEG is recorded at a higher sampling rate, such as 500 Hz, it may be downsampled to 250 Hz after filtering. Downsampling is performed only after filtering to avoid aliasing, where high-frequency noise could appear as false lower-frequency activity after reducing the sampling rate.

4.2 Artifact Handling

Artifact handling is especially important in this task because speech-related EEG recordings can easily be contaminated by non-brain activity. Although the positive window ends at speech onset, the period immediately before speaking may still contain artifacts from preparation of the mouth, face, or body.

Possible artifacts include eye blinks, eye movements, jaw movement, tongue movement, facial muscle activity, swallowing, electrode movement, and poor electrode contact. These artifacts can produce large voltage changes or abnormal signal fluctuations that may not reflect true cortical activity.

Artifact handling will be based mainly on simple rejection criteria and visual inspection. EEG windows will be rejected if the amplitude is extremely large, for example exceeding a conservative threshold such as $\pm 100\text{--}\pm 150\text{ }\mu\text{V}$, or if the signal variance is abnormal relative to the participant’s baseline recording. Signal variance measures how strongly the EEG amplitude fluctuates within a window; unusually high variance may indicate muscle activity, electrode movement, or other artifacts.

A short resting baseline will be recorded for each participant to estimate typical EEG amplitude and variance. Windows with variance far above the participant’s normal range, such as values above the 95th or 99th percentile, may be rejected. Trials with poor electrode contact or obvious recording problems will also be removed. In addition, a subset of trials will be visually inspected to confirm that the automatic rejection rules are reasonable.

Post-speech windows will not be used as positive intent samples because overt speech produces strong jaw, tongue, and facial muscle artifacts. The positive window will end at the detected speech onset to reduce contamination from actual speech movement.

4.3 Window Extraction

For each detected participant speech onset time t_0 , the positive EEG sample will be extracted as:

$$X^+ = \text{EEG}[t_0 - 500\text{ ms}, t_0]$$

This gives the 500 ms interval immediately before speech begins. At a sampling rate of 250 Hz, each positive window contains:

$$0.5 \text{ s} \times 250 \text{ Hz} = 125$$

time points per channel. With four electrodes, each input window has shape:

$$4 \times 125$$

Negative samples will be extracted from idle, silent, and listening periods, as well as from windows that do not end at the participant’s own speech onset.

Finally, each EEG channel will be normalized before modeling. Normalization parameters, such as the mean and standard deviation, will be computed using only the training set and then applied to the validation and test sets to avoid data leakage.

5 Modeling

The task is formulated as a binary classification problem. Each input sample corresponds to one 500 ms EEG window, and the model predicts whether the window represents intent-to-speak or no intent:

$$y = \begin{cases} 1, & \text{intent to speak} \\ 0, & \text{no intent to speak} \end{cases}$$

The primary model will be a Support Vector Machine (SVM) classifier. An SVM is chosen because it is suitable for small to medium-sized datasets and can work well with low-dimensional feature vectors. This is appropriate for the present task because only 2–4 EEG electrodes are used, and the amount of available training data is expected to be limited compared with large-scale deep learning settings.

5.1 Input Representation

At an effective sampling rate of 250 Hz, each 500 ms EEG window contains 125 time points per channel:

$$0.5 \text{ s} \times 250 \text{ Hz} = 125 \text{ samples}$$

With four electrodes, the raw EEG window has shape:

$$4 \times 125$$

Instead of giving the raw EEG matrix directly to the SVM, each window will be converted into a feature vector. Features will be extracted separately from each electrode. The feature set will include both time-domain and frequency-domain information.

Time-domain features may include:

mean amplitude, variance, linear slope

The mean amplitude captures the average voltage level in the window. Variance captures how strongly the signal fluctuates. The slope captures slow trends in the signal that may be related to readiness or preparation activity.

Frequency-domain features will include band power in EEG frequency ranges that are relevant to speech and motor preparation:

$$\begin{aligned} \theta &: 4 - 8 \text{ Hz} \\ \mu/\alpha &: 8 - 13 \text{ Hz} \\ \beta &: 13 - 30 \text{ Hz} \end{aligned}$$

Band power measures how much signal energy is present in each frequency band. These features are included because intent-to-speak may be associated with changes in sensorimotor rhythms and motor-preparation activity over fronto-central and central electrodes.

For example, if six features are extracted from each of four electrodes, the final input to the SVM will be a 24-dimensional feature vector:

$$4 \text{ electrodes} \times 6 \text{ features per electrode} = 24 \text{ features}$$

Thus, each EEG window is represented as:

$$X = [x_1, x_2, \dots, x_{24}]$$

where the elements of X summarize the temporal and spectral properties of the 500 ms EEG window.

5.2 Why This Model Should Work

The proposed electrode locations are focused on fronto-central and central brain regions, which are relevant for speech planning and motor preparation. Since speaking requires preparation of the lips, tongue, jaw, vocal tract, and breathing muscles, EEG activity before speech onset may differ from EEG activity during idle or listening periods.

The SVM will attempt to learn a decision boundary between feature vectors from pre-speech windows and feature vectors from no-intent windows. Since EEG is noisy and the number of electrodes is small, a feature-based SVM is more realistic than a large deep neural network. The model is also relatively interpretable, since the extracted features can be inspected to understand whether time-domain or frequency-domain information contributes more to classification.

6 Training and Loss

6.1 Training Setup

Each sample in the dataset will be one EEG window with a binary label. Positive samples will be extracted from the interval:

$$[t_0 - 500 \text{ ms}, t_0]$$

where t_0 is the participant’s speech onset time. Negative samples will be extracted from idle, silent, and listening periods, as well as from windows that do not end at the participant’s own speech onset.

Two evaluation settings will be considered: within-subject and cross-subject.

In the within-subject setting, the model will be trained and tested separately for each participant. For each participant, trials will be divided into:

$$70\% \text{ training, } 15\% \text{ validation, } 15\% \text{ testing}$$

The split will be performed at the trial level rather than by randomly splitting neighboring windows. This avoids data leakage, since adjacent windows from the same trial may be highly similar.

In the cross-subject setting, the model will be trained on a group of participants and tested on unseen participants. For example, with 20 participants, approximately 14 subjects may be used for training, 3 for validation, and 3 for testing. This setting is more difficult but provides a better estimate of how well the system generalizes to new users.

6.2 Class Balance Strategy

No-intent windows are expected to be much more common than intent-to-speak windows. If all negative windows are used, the dataset may become severely imbalanced, and the model could achieve high accuracy by simply predicting no-intent most of the time. To avoid this, negative samples will be selected to obtain a controlled class ratio, such as:

$$1 : 1 \quad \text{or} \quad 1 : 2$$

for positive to negative samples. A 1:2 ratio allows the model to see more variety in no-intent examples while still preventing the negative class from dominating training. Class weights may also be used if needed.

6.3 Loss Function

Since the primary model is an SVM classifier, the training objective will be hinge loss. Hinge loss encourages the classifier not only to classify samples correctly, but also to classify them with a sufficient margin from the decision boundary.

For an SVM, labels are commonly represented as:

$$y \in \{-1, +1\}$$

where $+1$ represents intent-to-speak and -1 represents no intent. If the model output score for a sample is $f(x)$, the hinge loss is:

$$L = \max(0, 1 - yf(x))$$

If a sample is correctly classified with a large enough margin, the loss becomes zero. If the sample is misclassified or lies too close to the decision boundary, the loss is positive. This is useful for EEG because the signal is noisy, and a larger margin may improve robustness.

6.4 Evaluation Metrics

Performance will be evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and a confusion matrix.

Accuracy measures the overall fraction of correctly classified samples. However, accuracy alone is not sufficient because no-intent samples may be more common than intent-to-speak samples.

Precision measures how many predicted intent-to-speak samples are actually correct:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall measures how many true intent-to-speak samples are detected:

$$\text{Recall} = \frac{TP}{TP + FN}$$

The F1-score combines precision and recall:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

F1-score will be emphasized because it balances missed detections and false alarms. ROC-AUC will also be reported to measure how well the model separates intent and no-intent windows across different decision thresholds.

7 Expected Performance

With only 2–4 EEG electrodes, the expected performance should be considered moderate rather than near-perfect. The proposed electrodes cover fronto-central and central regions related to speech planning and motor preparation, but they still provide limited spatial information compared with a high-density EEG system. In addition, EEG signals are weak and noisy, and speech-related recordings may be affected by eye movements, jaw movement, facial muscle activity, and subject-to-subject variability.

7.1 Within-Subject Performance

Within-subject performance is expected to be higher because the model is trained and tested on different trials from the same participant. EEG patterns, baseline rhythms, electrode contact, and speaking style are more consistent within the same individual. For controlled tasks such as read-aloud and question-answer conditions, a realistic within-subject accuracy is expected to be approximately:

$$65\% - 80\%$$

Performance may be higher for clean controlled trials and lower for more natural conversation trials.

7.2 Cross-Subject Performance

Cross-subject performance is expected to be lower because the model must generalize to participants not seen during training. EEG signals vary across people due to differences in head anatomy, baseline rhythms, electrode contact, and speaking behavior. Therefore, realistic cross-subject accuracy is expected to be approximately:

$$55\% - 70\%$$

In naturalistic or meeting-like settings, performance may be closer to the lower end of this range. Even modest above-chance performance can still be meaningful because the task is difficult with only a few EEG electrodes.

7.3 Detection Latency

Although positive samples are defined using the full 500 ms window before speech onset, the intent-to-speak signal is not expected to be equally strong throughout the entire interval. Detection is likely to be more reliable closer to speech onset, when motor preparation for articulation is stronger.

The system is expected to detect intent most reliably around:

200 – –300 ms before speech onset

Detection as early as 500 ms before speech onset may be possible in some trials, but it is expected to be less reliable. Therefore, performance should be reported not only as classification accuracy, but also in terms of how early before speech onset reliable predictions can be made.